YATHARTH GOYAL

APPLIED AI · COMPUTER VISION · ML ENGINEERING

Jaipur, Rajasthan, India | yatharthgoyal657@gmail.com | GitHub | LinkedIn | LeetCode | Portfolio code

TECHNICAL SKILLS

Languages Java, Python, JavaScript, SQL, HTML, CSS
Frameworks & Libraries FastAPI, scikit-learn, React, Express.js, REST APIs
Databases PostgreSQL, PostGIS, SQLite, Redis / BullMQ
Cloud & Infrastructure Docker, Render, Neon, AWS SAM (deployment templates)
Developer Tools Git, GitHub Actions, pytest, Playwright; Java DSA: 16-topic repository
Machine Learning & AI YOLO, computer vision evaluation, pandas, NumPy, SciPy / HiGHS, MILP

SELECTED PROJECTS & ENGINEERING CONTRIBUTIONS

RouteGuard 2.0 | Road-Hazard Intelligence · IN PROGRESS

Python · YOLO · React/Leaflet · Express · PostGIS · TomTom

- Evaluating YOLO road-damage models with dataset-integrity checks, empty-label reviews, split-leakage audits and reproducible experiments.
- Integrating experimental computer vision with PostGIS-backed hazard intelligence and route comparison; research target ≥95% severe-hazard precision and recall remains unmet.

AegisED | Emergency Operations & Decision Support · LIVE DEMO | SOURCE | DEMO

React · Express · PostgreSQL · FastAPI · scikit-learn · Render

- Integrated a FastAPI/scikit-learn triage-support service into a React/Express emergency operations platform; recommendations require human review and allow recorded overrides.
- Compared four candidate models and evaluated on 900 held-out synthetic scenarios: macro-F1 0.789; these results are not clinical validation.
- Implemented PostgreSQL-backed workflows with row-level locking, authorization and auditable patient/resource transitions.

EvalForge | Evidence-Based Assessment Intelligence · PRIVATE · NOT DEPLOYED

React · Express · PostgreSQL · Redis/BullMQ · FastAPI · Docker

- Built a rubric-based evaluation workflow that anchors positive scores to exact response evidence spans and keeps instructor review in control.
- Separated Express orchestration, PostgreSQL durable jobs, BullMQ workers and the Python/FastAPI evaluator; guarded invalid or incomplete evaluator outputs.
- Documented 15/15 backend tests and 7/7 evaluator tests locally; no public deployment or externally validated scoring accuracy is claimed.

Grid Flex | Industrial Energy Scheduling Optimizer · LOCAL PROTOTYPE | SOURCE

Python · SciPy/HiGHS · MILP · JavaScript · AWS SAM

- Modeled synthetic industrial energy scheduling as a mixed-integer linear program using SciPy/HiGHS across cost, balanced and carbon objectives.
- Developed independent constraint verification and deterministic solar-shortfall, grid-outage and battery-wear sensitivity analyses.
- Documented 52/52 passing local regression tests; results are modeled scenarios, not measured emissions or deployed AWS outcomes.

EDUCATION & CERTIFICATIONS

Bachelor of Technology (B.Tech), Information Technology | JECRC, Jaipur | Expected 2028
CGPA: 9.00/10.00
Certifications Credentials withheld pending issuer-issued verification; no exam passes claimed.